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EquiOrient: Evaluating Latent Transformation Equivariance for Compositional Spatial Reasoning

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Abstract

Can training a VLM's answer-relevant spatial latent to obey a predeclared geometric transformation algebra improve compositional generalization beyond matched augmentation and output-consistency baselines? We test this on the dihedral group D_4 (horizontal reflection H and $90°$ rotation R, with five held-out compositions including noncommutative pairs like RH), using 8-way compass-direction labels on synthetic scenes with dense homogeneous clutter. We compare six matched arms differing only in structural objective, under a fully deterministic protocol (seeded initialization, SHA-256-seeded data noise, provenance-annotated results). EquiOrient achieves strong, specific latent algebra compliance: the structural loss is nonzero and convergent, and the wrong-geometry arm learns its own incorrect law. Behaviorally, at the scarce-data regime ($N{=}128$, ten matched seeds) the correct law is significantly distinguishable from the wrong law ($+0.76$pp, 95% CI $[+0.5, +1.0]$pp, $p < 0.001$, 10/10 seeds) but not from plain augmentation (-1.4pp, CI $[-3.0, +0.4]$pp, n.s.). At $N \geq 512$ every arm reaches ceiling (≥ 0.999) because the VLM backbone's region-pooled features are already robust to D_4 transforms. The result is a precisely quantified mixed outcome: correct geometric structure provides a small, reliable low-data advantage over incorrect structure, but no measurable advantage over unconstrained augmentation, and no behavioral differentiation once the backbone saturates the task.

1. Introduction

Spatial reasoning in vision–language models (VLMs) often requires compositional generalization: predicting properties of object relationships under transformations not seen during training. A natural hypothesis is that forcing a model's latent representation to obey the correct transformation algebra should improve this generalization. We test this hypothesis rigorously.

Phase 1 of EquiOrient [?] established that a manipulation-verified equivariance objective can train a VLM's answer-path latent to obey a predeclared transformation algebra with strong specificity. However, that evaluation had critical limitations: a binary left/right task where the answer algebra is closed, one seed, and a composition test ($V \circ H$) where the wrong law coincidentally matched the correct one.

This paper presents Phase 2, which fixes every identified limitation:

- Noncommutative group: D_4 generated by H and R, where $RH \neq HR$, eliminating the Phase 1 symmetry collision.
- 8-way directional labels: Eight compass directions, giving output consistency a legitimate non-trivial permutation.
- Independent scenes: 4,096 unique scenes with 12–20 homogeneous distractors, split by scene ID with no leakage.
- Five seeds × six arms: 30 confirmatory runs for tight confidence intervals.
- Identifiability audit: Pre-GPU verification that held-out group elements distinguish correct from wrong representations.
- Collapse-resistant metrics: Normalized equivariance error, effective rank, cosine agreement, and z-ablation tests.

Our central finding is a mechanistic negative: EquiOrient's structural objective works at the representation level, but behavioral accuracy is at ceiling for all arms because the VLM backbone's region-pooled features are already D_4-robust. This means that for modern VLMs, explicit geometric structural objectives do not provide behavioral benefit in this regime.

2. Related Work

Latent group representations. The Homomorphism AutoEncoder [4] learns group-structured latents from

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observed transitions. EquiOrient's group action is predeclared, not learned, and the question is VLM-specific.

Representation-level geometric supervision. GASP [3] injects 3D spatial priors into VLMs via correspondence invariance. EquiOrient asks whether relation-relevant components should transform non-uniformly, with a wrong-action causal control.

Transformation-pair consistency. SAGE [5] trains VLMs for logical consistency under paired operations. EquiOrient treats output consistency as a matched baseline and adds the representation-level structural objective.

Equivariant representation learning. Cohen & Welling [2] and Bronstein et al. [1] are foundational. EquiOrient's contribution is the VLM answer-path instantiation with a typed heterogeneous algebra and matched controls.

3. Method

3.1. D4 algebra and representation

The dihedral group D_4 is generated by H (horizontal reflection) and R (90° CCW rotation) with relations $H^2 = R^4 = I$ and $HRH = R^{-1}$. In image coordinates (x rightward, y upward):

$$H = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}, \quad R = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

The eight elements are $\{I, R, R^2, R^3, H, RH, R^2H, R^3H\}$. Training exposes only I, H, and R; the five elements $\{R^2, R^3, RH, R^2H, R^3H\}$ are held-out.

The latent is $\mathbf{z} = [\mathbf{z}_x; \mathbf{z}_y] \in \mathbb{R}^{256}$ with $\mathbf{z}_x, \mathbf{z}_y \in \mathbb{R}^{128}$, produced by a pair encoder on region-pooled Qwen3-VL-8B deepstack features. The correct representation is $\rho(g) = G_g \otimes I_{128}$, giving:

$$\rho(H)[\mathbf{z}_x, \mathbf{z}_y] = [-\mathbf{z}_x, \mathbf{z}_y], \quad \rho(R)[\mathbf{z}_x, \mathbf{z}_y] = [-\mathbf{z}_y, \mathbf{z}_x].$$

The wrong representation uses $\tilde{\rho}(R) = R_{180} = -I$ and $\tilde{\rho}(H) = H$. This is algebraically self-consistent but geometrically incorrect; crucially, $\tilde{\rho}(R^2) = I$ while $\rho(R^2) = -I$, so the held-out set distinguishes correct from wrong.

3.2. Identifiability audit

Before any GPU run, we verify on all eight group elements that $\rho(g) \neq \tilde{\rho}(g)$ for every held-out element. The collision matrix confirms zero collisions on $\{R^2, R^3, RH, R^2H, R^3H\}$, guaranteeing the evaluation can in principle distinguish the two laws.

3.3. Eight-way directional task

Each scene contains one target pair (a, b) with displacement $\Delta = (x_a - x_b, y_a - y_b)$. The label is the compass direction of Δ among eight classes: right, upper-right, above, upper-left, left, lower-left, below, lower-right. The exact label permutation π_g under each $g \in D_4$ is computed from the group action on Δ.

3.4. Six matched arms

All arms share: Qwen3-VL-8B-Instruct (frozen text backbone), vision LoRA (rank 16, qkv/proj/c_fc/c_proj), PairEncoderV2 (8192→512 →256), RelationHead8 (256→8), AdamW (lr 10^{-4}, batch 8, 2 epochs). The only difference is the structural objective:

Arm	Answer loss	Structural term
Original SFT	identity only	—
Augmentation	identity + transform	—
Output consistency	identity + transform	$\mathrm{MSE}(\ell_{Tx}, \pi_g(\ell_x))$
Latent invariance	identity + transform	$\|\mathbf{z}(x) - \mathbf{z}(Tx)\|^2$
EquiOrient	identity + transform	$\|\rho(T)\mathbf{z}(x) - \mathbf{z}(Tx$
Wrong geometry	identity + transform	$\|\tilde{\rho}(T)\mathbf{z}(x) - \mathbf{z}(Tx$

Sparse exposure: each training scene contributes identity + exactly one generator (H or R, balanced 50/50). A manipulation check asserts structural loss $> 10^{-6}$ every epoch.

3.5. Dataset

4,096 independent scenes (512 dev, 2048 train pool, 512 val, 1024 test). Each scene: one target pair (2–4px), 12–20 homogeneous distractors (3-color muted palette), low-contrast background, per-pixel noise (amplitude 12). Balanced 8-way labels. Scene IDs disjoint across splits. Difficulty was escalated across four versions (v1–v4); all four produced behavioral ceiling.

4. Verification Protocol

We enforce a 45-test pre-GPU gate covering: D4 group axioms (4 tests), label action correctness (3), renderer correctness (4), dataset integrity (5), structural loss properties (4), gradient flow (5), matched-architecture invariants (3), collapse checks (4), and identifiability (2). All 45 must pass before any GPU run. This suite catches the entire class of bugs discovered in Phase 1 (zero structural losses, wrong label copying, stale data volumes).

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Arm	Unseen acc.	Struct. loss
Original SFT	0.9994 ± 0.0007	0.000
Augmentation	1.0000 ± 0.0000	0.000
Output consistency	0.9992 ± 0.0014	0.459
Latent invariance	0.9998 ± 0.0004	0.224
EquiOrient	1.0000 ± 0.0000	0.112
Wrong geometry	1.0000 ± 0.0000	0.176

Table 1. Unseen-group accuracy and structural loss (mean ± std over five seeds). All arms are at ceiling.

Arm	I	R	R2	R3	H	RH	R2H	R3H
Aug.	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Eq.	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Wrong	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Inv.	0.999	1.000	1.000	1.000	0.999	0.999	1.000	1.000
SFT	0.999	1.000	1.000	1.000	1.000	0.999	0.999	0.999

Table 2. Per-transform accuracy (5-seed mean). All arms at ceiling.

N	Aug.	Eq.	Wrong	Eq−Aug (CI
128	0.955 ± 0.023	0.941 ± 0.017	0.934 ± 0.017	-1.4pp [-3.0, +
512	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000	0.0pp
2048	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000	0.0pp

Table 3. Unseen-group accuracy (mean ± std). N=128: 10 matched deterministic seeds. N=512: 30 runs. N=2048: 20 runs.

5. Results

5.1. Confirmatory results (5 seeds)

1 reports the confirmatory results across five seeds ($N = 512$, $\lambda = 1.0$).

Behavioral ceiling. Every arm achieves $\geq$ 0.999 unseen-group accuracy with overlapping confidence intervals. Per-transform accuracy is ≥ 0.998 for all eight D_4 elements across all arms (Table 2). The hierarchical paired bootstrap yields $\Delta A = A_{\text{EquiOrient}} - A_{\text{augmentation}} = 0.0000$ with 95% CI $[0.0000, 0.0000]$, ruling out a $\geq$ 3pp benefit.

Latent equivariance. EquiOrient's structural loss is 0.112 ± 0.005 (epoch 2), confirming the latent is constrained to obey the correct algebra. The wrong-geometry arm's structural loss is 0.176 ± 0.010—it learns its own (incorrect) law. Output consistency has higher structural loss (0.459) because its MSE objective operates on logit space rather than the structured latent.

Training dynamics. EquiOrient's answer loss converges from $4.106 \rightarrow 0.211$ across two epochs, while its structural loss decreases from $0.318 \rightarrow 0.108$. The wrong-geometry arm shows analogous convergence ($4.424 \rightarrow 0.388$ answer, $0.351 \rightarrow 0.174$ structural). Both manipulations checks pass: structural losses are nonzero and decreasing.

Why the ceiling exists. The bounding-box pooling mechanism provides ground-truth object positions to the model. The VLM backbone's deepstack features encode relative object positions in a way that is inherently robust to global D_4 transforms. Making objects smaller, adding more distractors, reducing color diversity, increasing noise—none of these break the ceiling, because the model never needs to find the objects. The backbone's region-pooled spatial features are already D_4-equivariant by construction.

5.2. Sample efficiency (data-scale experiments)

To test whether the structural objective provides benefit when data is scarce, we trained four key arms (augmentation, output consistency, EquiOrient, wrong geometry) at $N = 128$ training scenes with ten matched seeds under a fully deterministic protocol (all random seeds fixed before model initialization; dataset noise seeded via SHA-256 of scene ID and transform, identical across processes). We additionally report $N = 512$ (30 runs) and $N = 2048$ (20 runs).

At N=128, no arm reaches ceiling. EquiOrient's paired contrast against augmentation is -1.4pp with 95% CI $[-3.0, +0.4]$pp (paired t-test $p = 0.16$, wins 3/10 seeds): there is no evidence that the correct geometric structure improves behavior beyond plain augmentation. However, the predeclared contrast against the wrong-geometry control is $+0.76$pp with 95% CI $[+0.5, +1.0]$pp ($p = 0.0003$, wins 10/10 seeds): the correct law is behaviorally distinguishable from the incorrect one even at N=128. At $N \geq 512$, all arms reach ceiling and every contrast is identically zero.

The picture is therefore mixed and precisely quantified: the correct structural law confers a small but reliable low-data advantage over the wrong law, but no advantage over an unconstrained augmentation baseline, and any differences vanish once the backbone saturates the task.

6. Discussion

The structural objective works; the task does not. EquiOrient successfully trains the latent to obey the correct algebra: the structural loss is nonzero and con-

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vergent ($0.318 \rightarrow 0.108$), and the wrong-geometry arm independently learns its own incorrect law ($0.351 \rightarrow 0.174$). The mechanism works. However, behavioral accuracy is at ceiling for all arms, because the bounding-box pooling provides ground-truth object positions to the model, eliminating the perceptual challenge that spatial compositionality requires.

Why visual difficulty does not help. We systematically escalated difficulty across four versions (v1–v4): reducing target size (6–9px $\rightarrow$ 2–4px), increasing distractor count (6–10 $\rightarrow$ 12–20), reducing color diversity (12 $\rightarrow$ 3 near-identical hues), and increasing per-pixel noise (amplitude 6 $\rightarrow$ 12). All four versions produced behavioral ceiling. The cause is architectural: the model pools features from grid cells specified by bounding boxes, so it never needs to locate objects or reason about their visual properties. The backbone's region-pooled spatial features encode relative positions in a way that is inherently robust to global D_4 transforms.

Implications for compositionality benchmarks. Our result reveals a design pitfall: benchmarks that provide bounding boxes as input may systematically overestimate spatial compositionality difficulty for modern VLMs. The backbone's visual features already encode the needed geometric information. Future benchmarks should consider: (1) natural images without bounding boxes, where object localization is part of the challenge; (2) relations requiring multi-step reasoning beyond single-pair displacement; (3) adversarial construction targeting known VLM failure modes; and (4) tasks where the answer cannot be inferred from a single pair's pixel coordinates.

What this paper contributes. The contribution is a precisely quantified mixed result with an architectural explanation. At scarce data (N=128, ten matched deterministic seeds), the correct geometric law is behaviorally distinguishable from the incorrect one (+0.76pp, $p < 0.001$, all ten seeds) but not from plain augmentation (−1.4pp, n.s.): structural constraints encode which law the latent obeys, yet provide no measurable advantage over an unconstrained baseline. At $N \geq 512$, bounding-box-pooled VLM features are already D_4-robust, so every arm reaches ceiling and all contrasts vanish. The finding is actionable: it tells the field that providing bounding boxes as input shortcuts the spatial reasoning challenge, and that future work on geometric structure in VLMs must evaluate without such shortcuts to test whether the correct law confers a genuine behavioral advantage.

7. Limitations

This study has several limitations that should guide future work:

- **Bounding-box shortcut.** The model receives ground-truth object positions, eliminating the perceptual challenge. The finding that behavioral accuracy is at ceiling is an artifact of this design choice, not a general statement about VLM spatial reasoning.
- **Frozen backbone.** The VLM text backbone is frozen; only the vision LoRA and downstream heads are trained. The structural objective cannot modify the backbone's spatial representations. Future work should test with an unfrozen backbone.
- **Synthetic data.** All scenes are synthetic plan-view renderings. The finding that VLM features are D_4-robust may not transfer to natural images where rotations introduce semantic artifacts.
- **Two epochs of training.** Each arm trains for only 2 epochs (∼128 steps). Longer training could reveal behavioral differences between arms, though the ceiling suggests otherwise.
- **Single group.** Only D_4 (8 elements) is tested. Larger groups (SO(2), SE(3)) may produce harder tasks where structural objectives provide benefit.
- **No comparison to prior methods.** SAGE [5] and GASP [3] are cited but not compared head-to-head.

8. Conclusion

We present EquiOrient Phase 2, a rigorous evaluation of latent transformation equivariance on the noncommutative dihedral group D_4, with 8-way labels, 30 confirmatory runs at N=512, ten matched deterministic seeds at N=128, and a 45-test verification suite. EquiOrient achieves strong, specific latent algebra compliance (structural loss 0.112, wrong-geometry 0.176). Behaviorally, the correct law is significantly distinguishable from the wrong law at N=128 (+0.76pp, $p < 0.001$) but not from plain augmentation (−1.4pp, n.s.), and at $N \geq 512$ all arms reach ceiling because bounding-box-pooled VLM features are already D_4-robust. The result is a precisely quantified mixed outcome with an architectural explanation: geometric structure encodes which law the latent obeys, but provides no measurable advantage once the backbone saturates the task. Future work must evaluate without bounding-box shortcuts to meaningfully test whether correct geometric structure confers a genuine behavioral advantage.

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